Seminar9

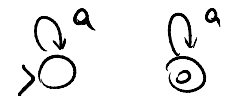
luni, 13 decembrie 2021 11:50

LR → AFD, AFN, ER, P.I.

LIC → AP, GIC, P.I.

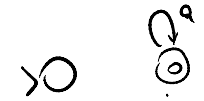
LR:

AFD: $M = (K, \Sigma, \delta, s_0, F)$ $\quad \delta(s_1, x) = s_2$



$\Sigma = \{a\}$

AFN: $M = (K, \Sigma, \Delta, s_0, F)$ $\quad (s_1, x, s_2)$



LIC:

APD: $M = (K, \Sigma, \Gamma, \Delta, s_0, F)$ $\quad ((s_1, x, \alpha), (s_2, \beta))$

GIC $\quad S \to aSb \mid SS$

$G = (V, \Sigma, R, S)$

$S \Rightarrow SS \Rightarrow aSbS$

ER: $\cup, \circ, *$ $\quad \emptyset, e$

LR: $\cup, \circ, *, \setminus, \cap, C$

$\forall L_1, L_2 \in LR \Rightarrow L_1 \cup L_2,\ L_1 \circ L_2,\ L_1 \cap L_2,$

$L_1^*,\ L_1 \setminus L_2,\ CL_1 \in LR.$

LIC: $\cup, \circ, ^*$

$\forall L_1, L_2 \in LIC \Rightarrow L_1 \cup L_2,\ L_1 \circ L_2,\ L_1^* \in LIC$

$\forall L_1 \in LIC$ și $L_2 \in LR \Rightarrow L_1 \cap L_2 \in LIC$

$L_1 \setminus L_2 \in LIC.$

$\forall L \in LR \Rightarrow L \in LIC$

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1. Fie $L = \{ w \in \{a,b\}^* \mid |w| = 2k+1,\ k \geq 0,$ și w are același simbol la început, mijloc și final $\}$.

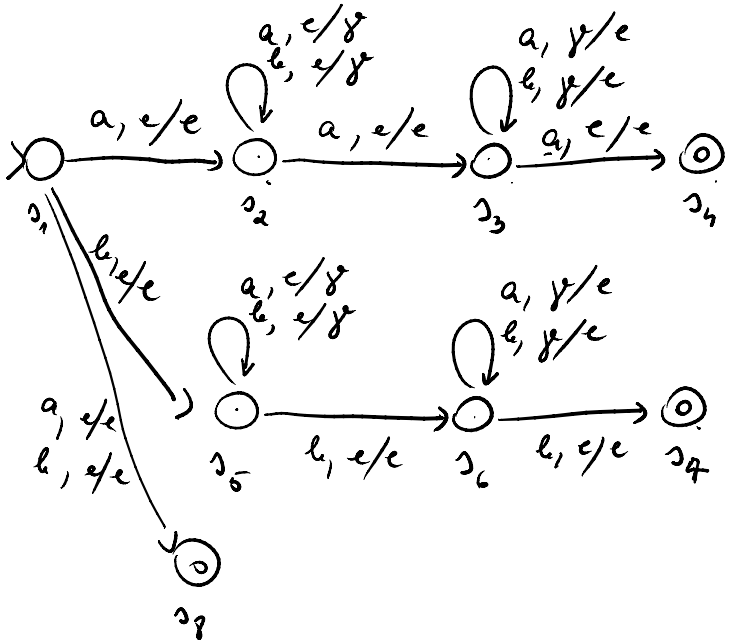
Cum este L în cel mai strict sens?

a). finit

b). LR

c). $\notin$ LIC

d) LIC



$S \to a S_1 a \mid b S_2 b \mid X$

$S_1 \to a \mid X S_1 X$

$S_2 \to b \mid x\, S_2\, x$

$x \to a \mid b$

2. Fie $E = a^* b^*$. Care din urm. exp. reg. sunt echivalente cu E?

a1. $(a \cup b)^* \varnothing^* b^* \equiv (a \cup b)^* b^*$

b1) $(a \cup \varnothing)^* (b \varnothing^* \cup a \varnothing)^*$

c1. $((a \cup b)^* \varnothing)\, a^* b^* \varnothing^* \equiv \varnothing\, a^* b^* e = \varnothing$

d1. $(a \cup b)^* \cap a^* b^* \equiv a^* b^*$

$L_1 \circ L_2 = \{ w \in \Sigma^* / w = w_1 w_2,$
$w_1 \in L_1$ și
$w_2 \in L_2 \}.$

$L_1 \circ \varnothing = \varnothing = \varnothing \circ L_1$

$\varnothing^* = e$

$(a \cup \varnothing)^* (b \varnothing^* \cup a \varnothing)^* \equiv a^* (b \cup \varnothing)^*$
$\equiv a^* b^*$

(a : $a \cup \varnothing$; $b \cdot e$: $b \varnothing^*$)

$(a \cup (b (c^*)))$

1. $*$
2. $\circ$
3. $\cup$

$a \cup (b c^*)$

3. Fie G o GIC:

$U \to XUX \mid S$

$S \to aTb \mid bTS$

$T \to XTX \mid X \mid e$

$X \to a \mid b$

a) $T \overset{*}{\Rightarrow} XX$ $\quad T \Rightarrow XTX \Rightarrow XX$

~~b)~~ $T \Rightarrow aba$

~~c)~~ $S \overset{*}{\Rightarrow} e$

~~d)~~ $X \overset{*}{\Rightarrow} aba$

4. Ce limbaj generează gramatica ?

$S \to ab \mid ba \mid SS \mid S \mid a \mid b \mid e$

~~a)~~ $\mathcal{L}((ab \cup ba \cup aa \cup bb)(a \cup b \cup e))^*)$

~~b)~~ $\mathcal{L}((aba \cup bab)^*)$

~~c)~~ $\mathcal{L}((aba \cup aab \cup baa \cup bab)^*)$

d) $\mathcal{L}((a \cup b)^*)$

$S \to SS$ $\quad S \Rightarrow SS \Rightarrow SSS \Rightarrow SSSS$

$(a \cup b)^* \equiv (a \cup b \cup e)^*$

$\mathcal{L}(((ab \cup ba \cup aa \cup bb)(a \cup b \cup e))^*)$

$((ab \cup ba \cup aa \cup bb)(a \cup b \cup e))^*$

$((ab \cup ba \cup aa \cup bb)(a \cup b \cup e))$

$\equiv ba\,e\,bb\,b = baabbb$

5.

6. Fie $L_1, L_2 \in LIC$ a.î. $L = L_1 \cup L_2$, $L \in LR$.

At.

a) nici L_1, nici L_2 nu trebuie să fie LR

b) L nu poate fi regulat

c) fie $L_1 \in LR$, fie $L_2 \in LR$

d) $L_1, L_2 \in LR$

$a^*b^* = a^*b^*$

a^*b^*

$L_1 = a^n b^m$, $n = m \in LIC$

$L_2 = a^n b^m$, $n \neq m \in LIC$

$L_1 \cup L_2 = a^*b^* \in LR$.

6. Fie Σ un alfabet și $L_1, L_2 \subseteq \Sigma^*$ a.î.

L_1 și $L_1 \cup L_2 \in LR$. At:

a) L_2 este întotdeauna LR

b) niciuna

c) L_2 poate să nu fie regulat

c) L_2 poate să nu fie regulat

~~d).~~ $L_2 = (L_1 \cup L_2) \setminus L_1$ $\quad L_1 \cap L_2 = \emptyset$

$\cup \ (L_1 \cap L_2)$

$L_1 = \{ a, ab \}$ $\quad L_1 \cup L_2 = \{ a, ab, aab \}$.

$L_2 = \{ aab, b \}$. $\quad (L_1 \cup L_2) \setminus L_1 = \{ aab \} \neq L_2$.

$L_1 = a^n b^m \in \underline{LR}$ $\quad a^* b^*$

$L_2 = a^n b^n \notin \underline{LR}$.

$L_1 \cup L_2 = a^n b^m \in \underline{LR}$

$a^* b^*$

7. Care e val. de adevăr a urm. afir.

i). $\forall M_1, M_2 \in \underline{AFD}$, $M_1 = (K_1, \Sigma, \underline{\delta_1}, s_1, F_1)$ și

$M_2 = (K_2, \Sigma, \delta_2, s_2, F_2)$, a.î. Σ^*

$F_1 = K_1$ și $F_2 = K_2 \Rightarrow \mathcal{L}(M_1) = \mathcal{L}(M_2)$ $\quad \Sigma^* \quad \Sigma^*$ A.

ii). $\forall M_1, M_2 \in AFN$, $M_1 = (K_1, \Sigma, \Delta_1, s_1, F_1)$ și

$M_2 = (K_2, \Sigma, \Delta_2, s_2, F_2)$, a.î.

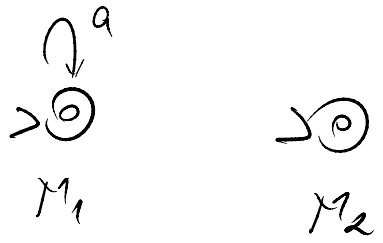
$F_1 = K_1$ și $F_2 = K_2 \Rightarrow \mathcal{L}(\underline{M_1}) = \mathcal{L}(\underline{M_2})$ F.

a). A, A

b). F, F

c). A, F

d). F, A



8. Fie $L = \{w \in \{a, b, c, d\}^* \mid \#_c(w) \% 2 = 0, \#_d(w) \% 3 = 1,$

$w = \underbrace{u v^2}\, v^*$

$w = \underbrace{uv}$

$\Rightarrow \#_a(w) = \#_b(w)$.

Cum este L în cel mai strict sens?

a). niciuna

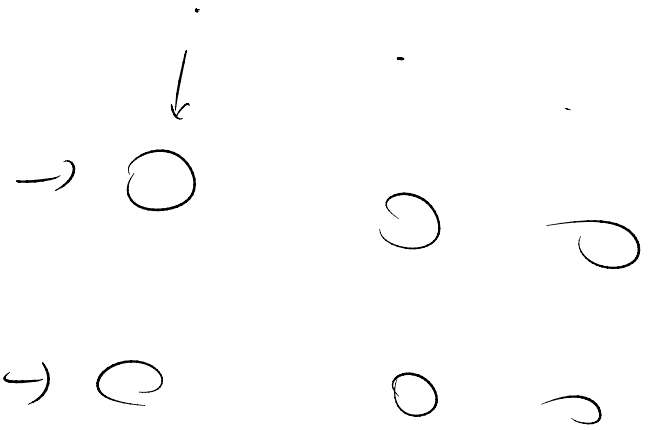
b). LR

c). LIC

d). finit.

$ab(cc)^*$

$ab(ccc)^*c$



$a^i b^j c^k$, $i < j < k$. $\notin L/C$.

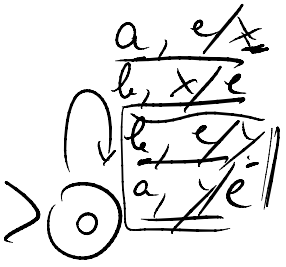
$a^n b^n c^n$, $n = 100$ $\in L_k$

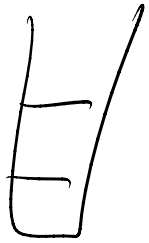
$\notin L/C$.

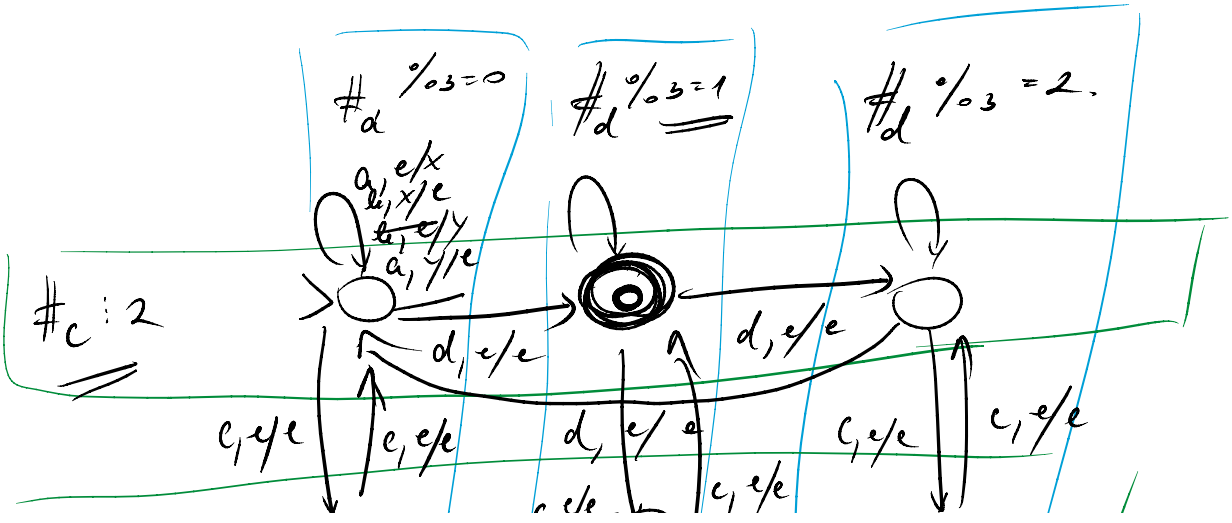
$\overline{L}$

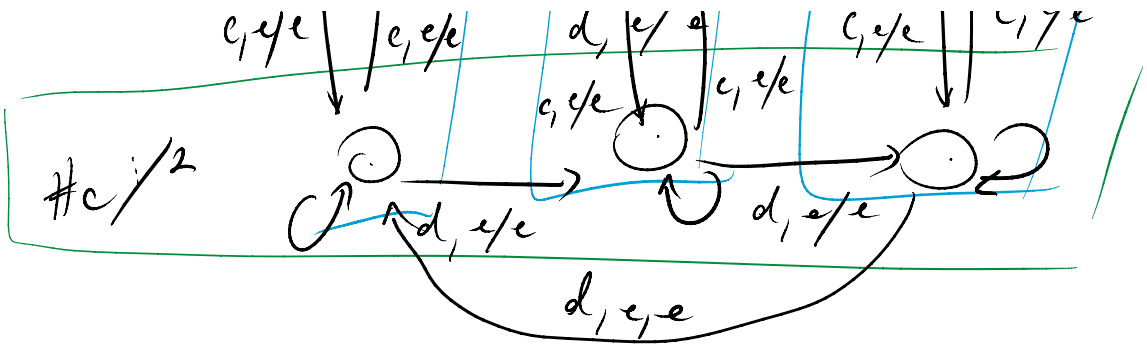
$a\,b\,b\,a$

$\#_a = \#_b$








#c/2
c, e/e
c, e/e
d, e/e
c, e/e
c, e/e
c, e/e
d, e/e
d, e/e
d, e, e